

Lab 02 Development Prompt History

File name: lab02_golloy_prompts.pdf

This document records the development history used for Lab 02: interpreting the activity, inspecting the dataset, planning calculations, coding numerical methods, debugging/checking likely issues, curve fitting, integration, dashboard organization, and final requirement checks. It is intended to represent the development process rather than only the final successful prompt.

1. Understanding the Activity

Prompt: I have a file of an activity. Can you help me identify the main things I need to calculate and the files I need to submit? I want to use the instructions as my checklist.

2. Identifying the Dataset

Prompt: this is the data needed

3. Numerical Differentiation

Prompt: can you help me check my approach for calculating the forward, backward, and centered first and second derivatives?

4. Curve Fitting and Model Selection

Prompt: I'm working on the curve-fitting part now. show me how to use `scipy.optimize.curve_fit` with the levenberg-marquardt method and explain what the initial guesses are for? also, help me compare their fit results and explain what I should consider when choosing one instead of simply choosing the model with the biggest R^2 ?

5. Numerical Integration and Cross-Check

Prompt: help me set up `scipy.integrate.quad` for the fitted function and explain how the result should be compared with `np.trapezoid` on the raw readings?

6. Dashboard Development and Python/JavaScript Separation

Prompt: i need to turn my python results into the dashboard required by the pdf. help me organize the sections and make sure the calculations stay in python instead of being done by javascript? before i finalize the files, can you help me check my script for common problems such as column names, timestamp conversion, curve-fitting errors, and incorrect units?

7. Reviewing Completed Outputs Against Lab Requirements

Prompt: with my completed outputs with the lab 02 requirements and tell me what is still missing or needs attention?

8. Required Prompt-History Deliverable

Prompt: give me the full prompt history in pdf: coding, debugging, modifying, explaining, and the prompts that produced the dashboard. The development history, not the last prompt that happened to work. name it lab02_golloy_prompts

Development Flow Summary

- Activity requirements and submission checklist were extracted from the Lab 02 activity instructions.
- The supplied reservoir-stage dataset was identified as the basis for the numerical analysis.
- The time-series workflow was developed around elapsed time in hours and depth in meters.
- Finite-difference formulas were reviewed for forward, centered, and backward first derivatives and for second-derivative handling.
- Candidate fitted models and the use of `scipy.optimize.curve_fit` with `method='lm'` were discussed.
- Initial parameter guesses (p_0), fit statistics, residuals, SSE, SST, R^2 , standard error, parameter uncertainty, t-statistics, and p-values were covered.
- Model selection was framed around residual behavior, physical interpretation, uncertainty, and error measures rather than R^2 alone.
- `scipy.integrate.quad` was set up to integrate the selected continuous fitted function, with `np.trapezoid` used as a cross-check on the raw readings.
- The dashboard was organized into an always-visible raw-data chart plus Derivatives, Fitted Curve/Statistics, and Area Under the Curve sections.
- Python was designated as the source of all numerical calculations, while HTML/JavaScript was limited to displaying precomputed results and rendering charts.
- Common implementation checks were identified for column names, timestamp conversion, curve-fitting warnings/errors, covariance results, missing data, and units.
- Final output requirements were reviewed against the Lab 02 instructions.